

# Coding Agent Assignment for Shortlisted Walk-in Candidates

## Coding Agent Assignment

### Objective

Build and deploy a small **AI Coding Agent** that can understand a codebase, accept a developer task, make or propose code changes, and show the result clearly.

### Problem Statement

Create a coding-agent application where a user can enter a coding request such as:

“Add input validation to this API and write a test for it.”

The agent should analyse the selected codebase or sample project, identify the relevant files, propose a plan, make changes or generate a patch, and present the output to the user.

### Minimum Requirements

#### 1. User Interface

- Provide either a simple web UI or CLI.
- The user should be able to enter a coding task in natural language.
- The app should show the agent’s plan before or while making changes.

#### 2. Codebase Handling

- Use a sample repository or a small project created by you.
- The agent should read multiple files from the project.
- It should identify which files are relevant to the requested task.

#### 3. Agent Behaviour

The agent should be able to:

- Understand the coding request.
- Inspect relevant files.
- Suggest or make code changes.
- Explain what was changed and why.
- Show the changed files or patch/diff.

## 4. Validation

- Add at least one meaningful test, validation step, or demo case.
- Include clear instructions for how to run the project locally.

## 5. Deployment

- Deploy the working application somewhere accessible online.
- The deployed link can be hosted on any suitable platform such as Render, Vercel, Netlify, Railway, Hugging Face Spaces, Streamlit Cloud, or any equivalent service.

## Submission Requirements

Candidates must share:

| Item                           | Required |
|--------------------------------|----------|
| Deployed application link      | Yes      |
| GitHub repository link         | Yes      |
| Short README                   | Yes      |
| Brief explanation of approach  | Yes      |
| Any assumptions or limitations | Yes      |

## Evaluation Criteria

| Area                     | What we will look for                                                            |
|--------------------------|----------------------------------------------------------------------------------|
| Agent design             | Clear flow from task understanding to file selection and code change             |
| Practical implementation | Working application, not just a concept                                          |
| Code quality             | Clean structure, readable code, and sensible error handling                      |
| AI usage                 | Meaningful use of LLM / agentic flow / tool-calling / retrieval where applicable |
| Deployment               | A usable deployed link that reviewers can open                                   |
| Communication            | Clear README and explanation                                                     |

## Notes for Candidates

- You may use any suitable tech stack.
- You may use any LLM provider or local model.
- Keep the scope small but complete.
- Focus on a working end-to-end flow rather than a large unfinished system.
- Do not include paid API keys or secrets in the repository.